

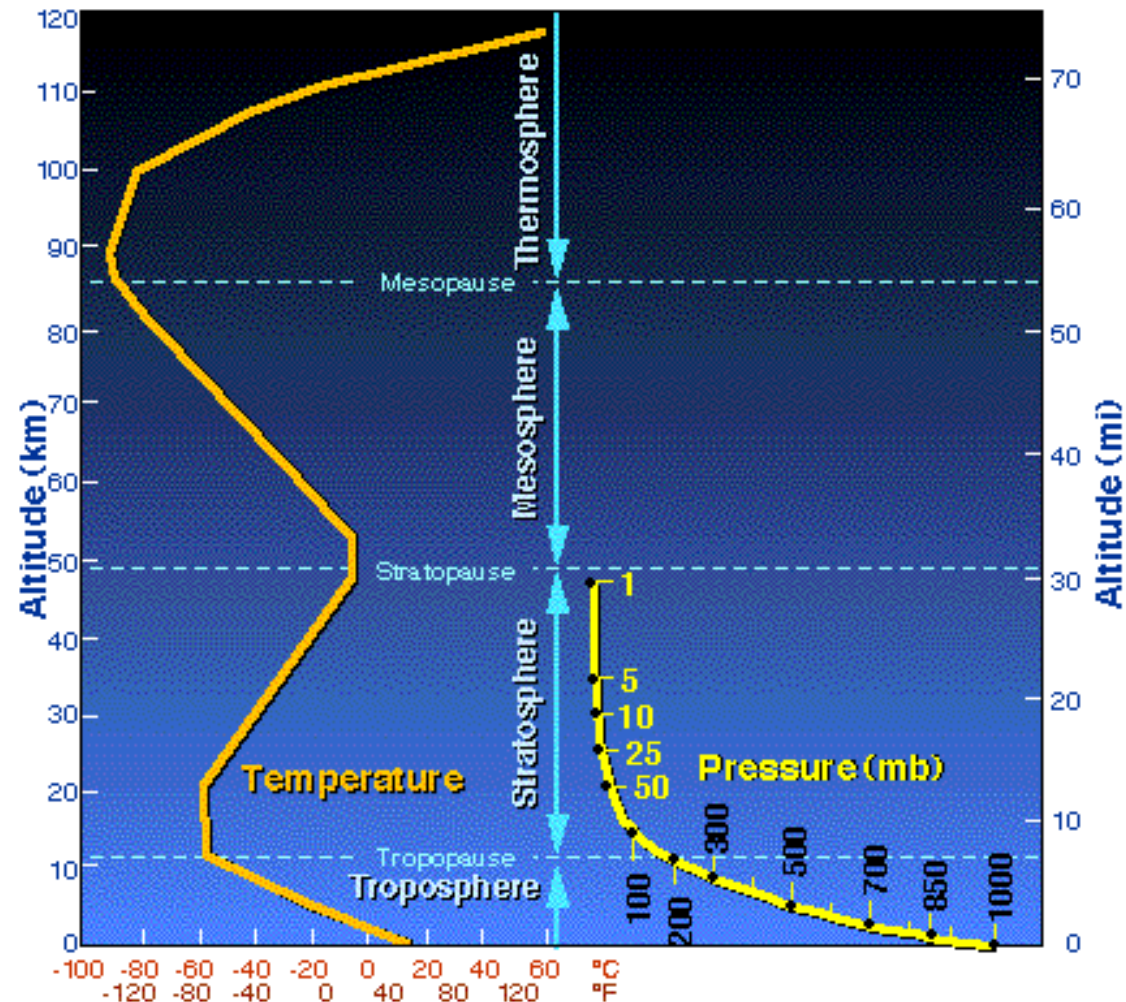
Introduction to Environmental Science EES102

Lecture by: Baerbel Sinha, IISERM

CLIMATE AND CLIMATE CHANGE

The earth's atmosphere is subdivided vertically into four major regions based on the air-temperature profile. The ozone layer that protects us from the sun's UV radiation is in the stratosphere.

Average atmospheric temperature and pressure in the atmosphere



WHAT IS CLIMATE

Climate is nothing but the long term average temperature, pressure and precipitation on Earth.

Weather is the short term variation of these variables

In a more local context we speak of “Climate zones” or the climate of a place. **The climate of a place actually defines what can grow where depending on temperature, and rainfall in relation to potential evapotranspiration.**

Tropical moist climate

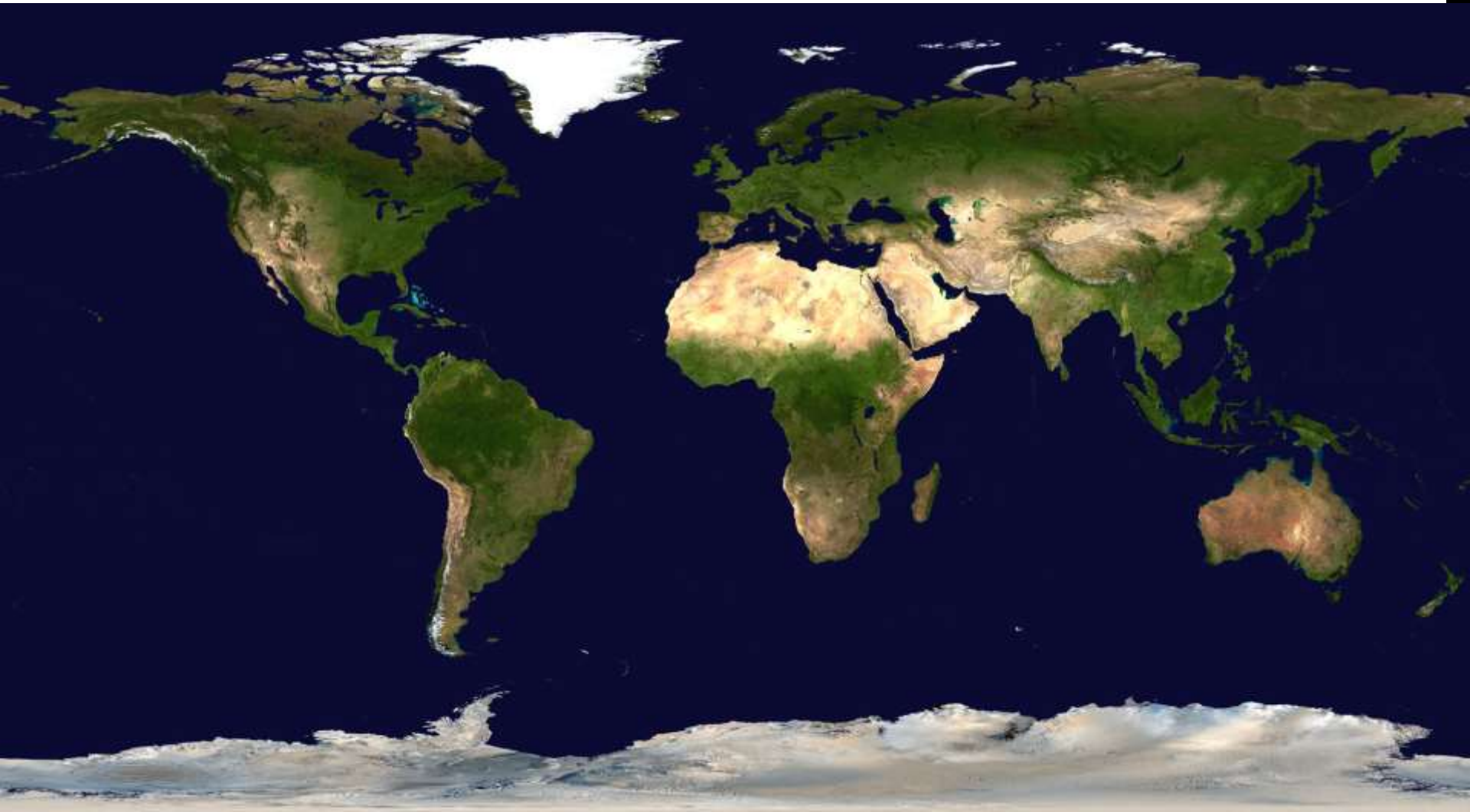


Arid climate

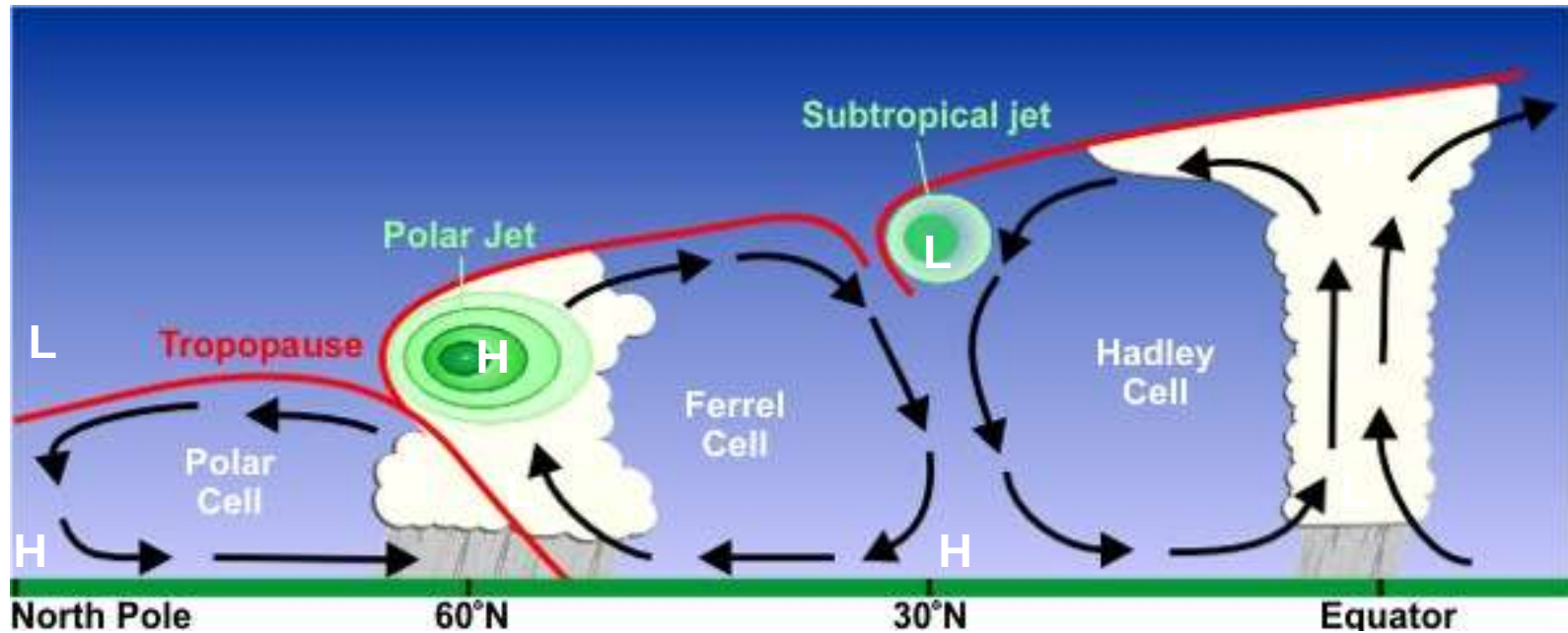


Temperate climate





VERTICAL STRUCTURE AND MECHANISMS

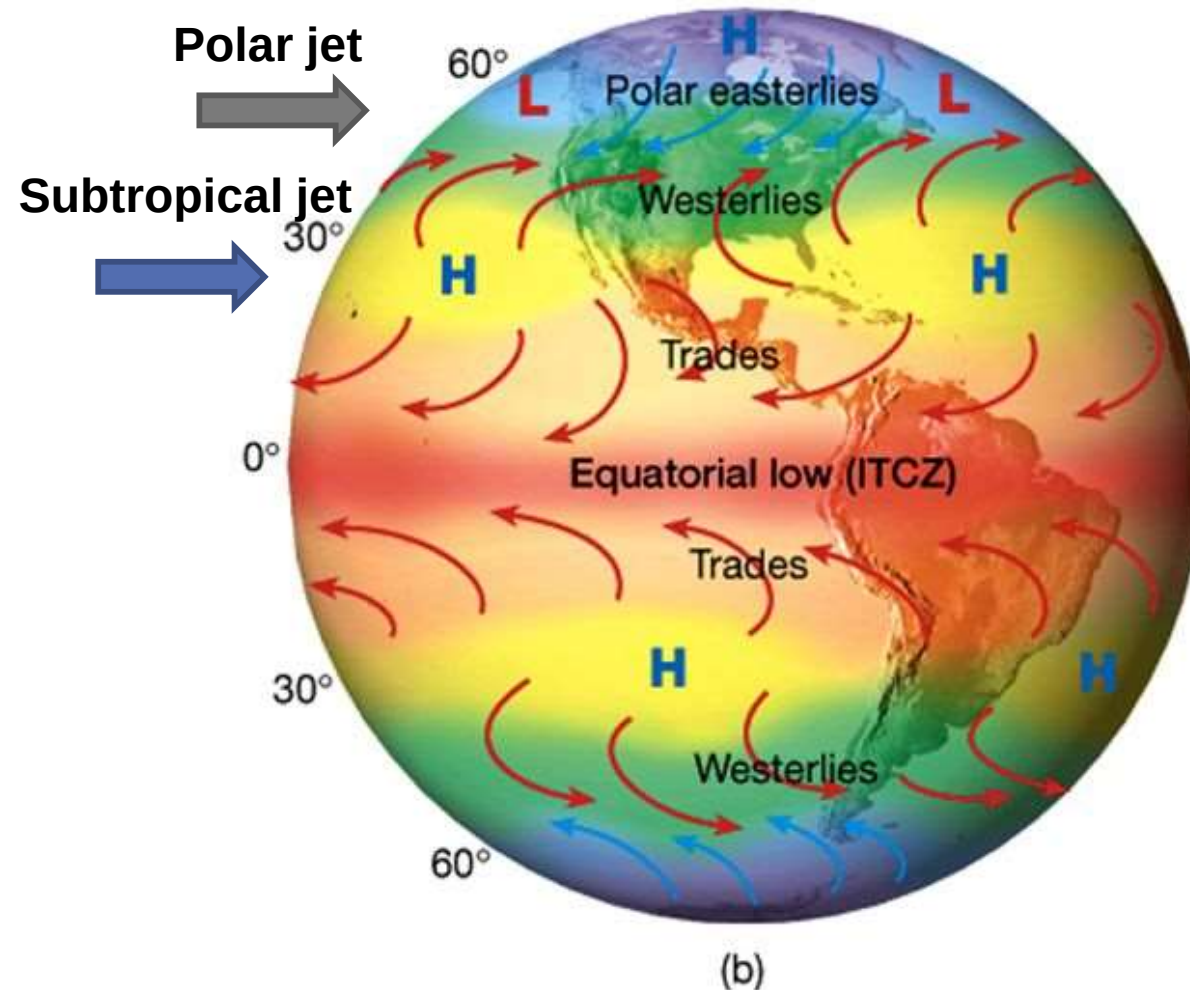


Polar Cell
(thermal):
Driven by heating
at 50 degree
latitude and
cooling at the
poles

Ferrel Cell
(dynamical):
Dynamical response
to Hadley and polar
cells

Hadley Cell (thermal):
*Heating in tropics →
forms surface low and
upper level high → air
converges equatorward
at surface, rises, and
diverges poleward aloft
→ descends in the
subtropics*

Semipermanent Pressure Systems



The green area is the zone where under normal conditions high pressure and low pressure systems alternate

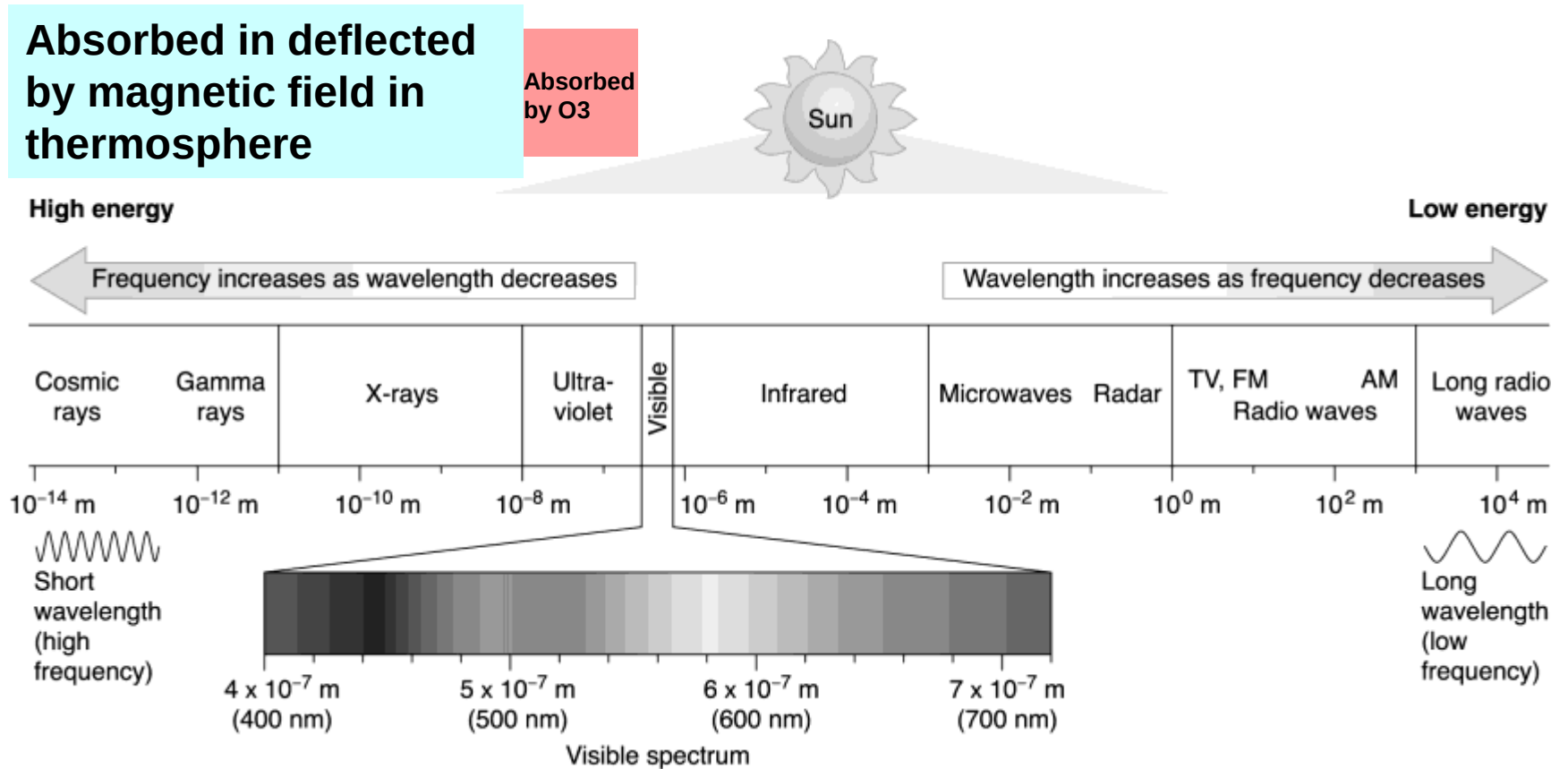
The real Earth with disruptions of the zonal pattern caused by large landmasses. These disruptions break up pressure zones into semi-permanent high and low pressure cells.

**WHERE ON EARTH IT IS HOT & COLD,
WET & DRY IS GOVERNED BY
CIRCULATION PATTERS**

**HOW HOT EARTH AS A WHOLE BODY
IS, IS GOVERNED BY THE
GREENHOUSE EFFECT**

**WHETHER AND HOW MUCH
TEMPERATURES GLOBAL AVERAGE
TEMPERATURES ARE CHANGING IS
GOVERNED BY THE RADIATION
BALANCE**

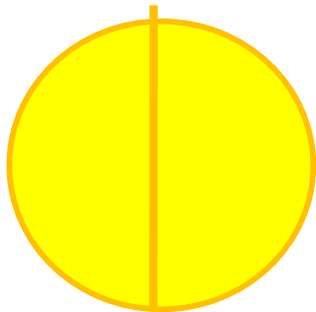
Figure 2.3 The electromagnetic spectrum



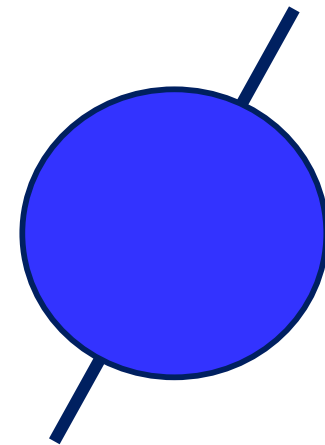
SOLAR FORCING

Radiative forcing (measured in watts per square meter) can be estimated in different ways for different components. For solar irradiance (i.e. "solar forcing"), the radiative forcing is simply the change in the average amount of solar energy absorbed per square meter of the Earth's area.

If we want to assess how much changes in solar intensity affect Earth considerations are as follows:



Only half of the sphere called Sun is radiation towards earth



Only half of the tilted sphere called earth is pointing towards sun at any point in time

RADIATIVE BALANCE OF THE EARTH

- In order to maintain a stable climate, the Earth must be in energetic eqb. between the radiation it receives from the Sun and the radiation it emits out to space.

- Total radiation flux emitted by the Sun ($T_s = 5800$ K) per unit time is given by the radiation flux σT_s^4 times the area of the Sun

$E_s = 4 \pi R_s^2 \sigma T_s^4$ where $R_s = 7 \times 10^5$ km is Sun's radius

- Earth is at distance “d” 1.5×10^8 km from the Sun; Solar radiation flux F_s at that distance is distributed uniformly over the sphere centered at the Sun of radius “d”

$F_s = 1370$ W/m²; F_s is called solar constant for Earth. Solar constants of other planets can also be calculated based on their distance from the Sun

SOLAR FORCING

The average annual solar radiation arriving at the top of the Earth's atmosphere is about 1361 W/m^2 . This represents the power per unit area of solar irradiance across the spherical surface surrounding the Sun with a radius equal to the distance to the Earth (1 AU). This means that the approximately circular disc of the Earth, as viewed from the Sun, receives a roughly stable $\sim 1361 \text{ W/m}^2$ at all times. The area of this circular disc is πr^2 , in which r is the radius of the Earth.

Because the Earth is approximately spherical, it has a total surface area of $4\pi r^2$. This means that the solar radiation arriving at the top of the atmosphere, averaged over the entire surface of the Earth, is simply divided by four to get $\sim 340 \text{ W/m}^2$. In other words, averaged over the year and the day, the Earth's atmosphere receives $\sim 340 \text{ W/m}^2$ from the Sun.

SOLAR FORCING

Since some radiation is reflected and only a fraction is absorbed by the surface, the insolation ($K_{\downarrow}$) must be multiplied by the fraction of incident sunlight that is absorbed, $F = (K_{\downarrow}) (1 - R)$, where R is the reflectivity (albedo)

Thus, the solar forcing is the change in the solar intensity divided by 4 and multiplied by 1-Albedo.

Likewise, a change in albedo will produce a solar forcing equal to the change in albedo divided by 4 multiplied by the solar constant.

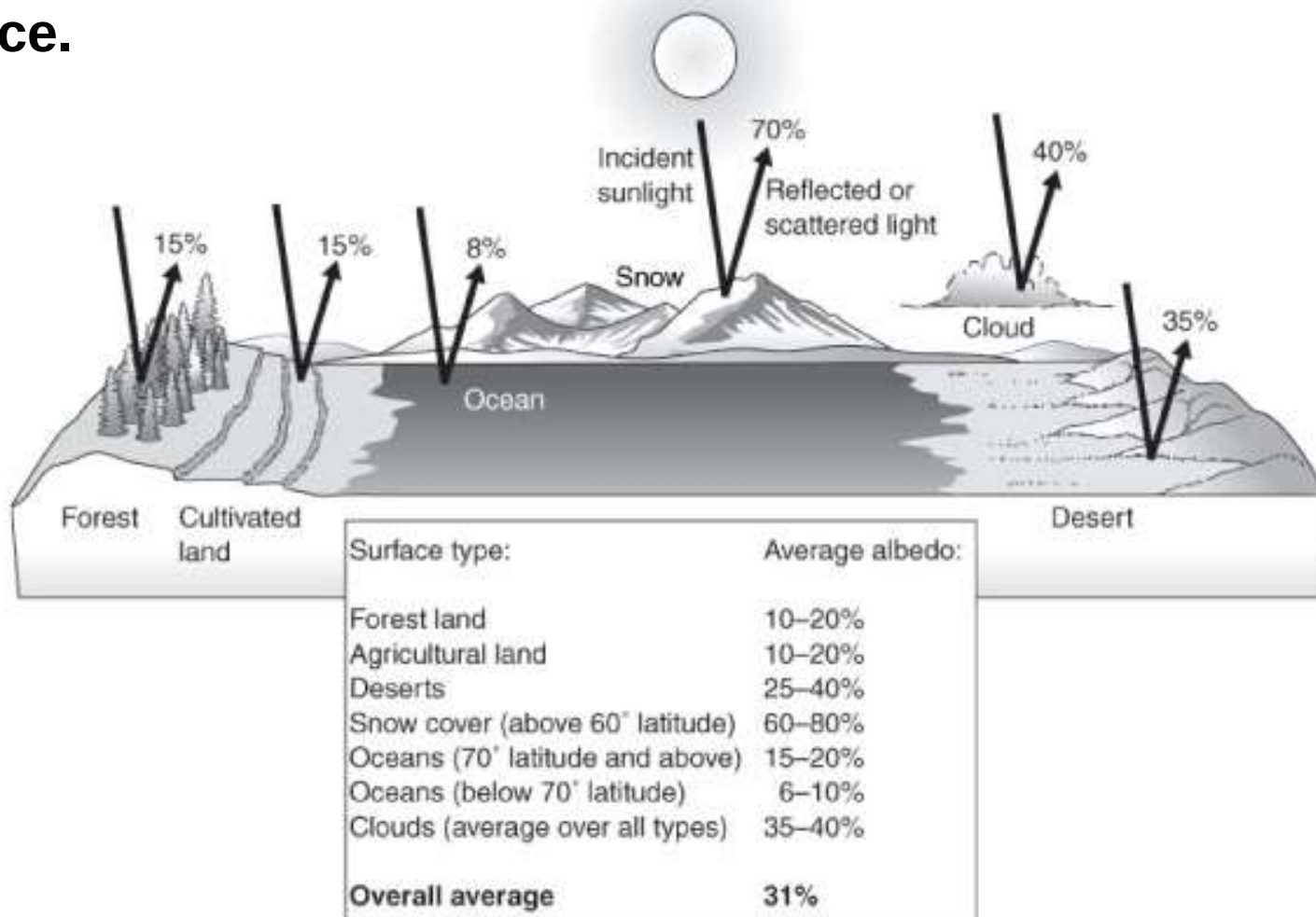
Solar forcing changes slowly and can be calculated from the Earth's orbital parameters (Astronomy) and change in sunspot activity. Albedo changes much faster.

The solar forcing is not the final radiative forcing acting on the earth system!

$$F_S \pi R_E^2 (1-A) / 4 \pi R_E^2 = F_S (1-A) / 4.$$

α = albedo, reflectivity of a surface (atmosphere or land)

Figure 2.5 Albedo varies depending on cloud cover and the type of surface.

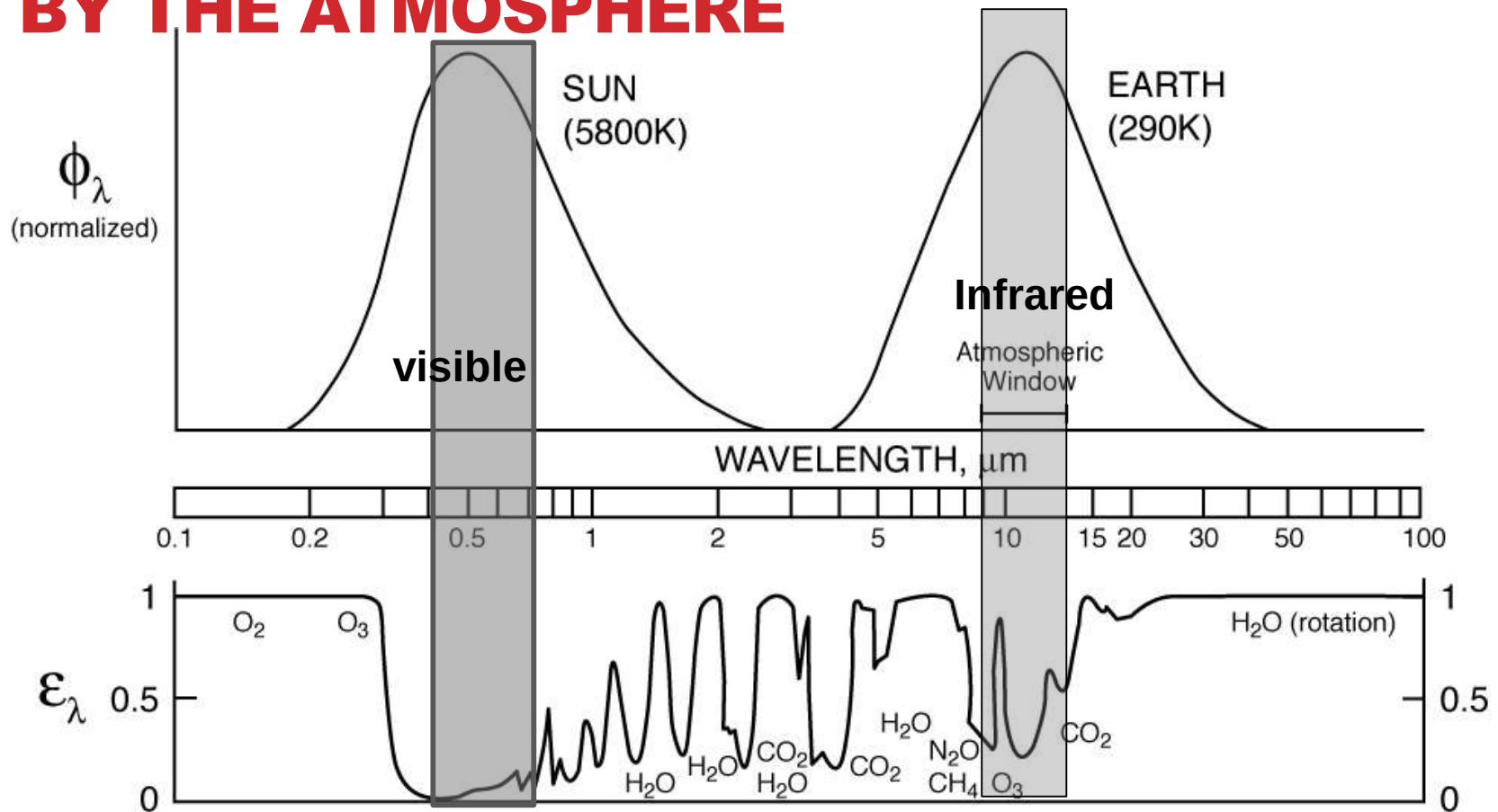


The albedo of the atmosphere varies with the amount of cloud cover and the concentration of *atmospheric aerosols* (solid dust and tiny liquid particles). The albedo of a surface is a material property related to the color of the material.

Table 2.3 Calculated and Actual Temperatures of the Surfaces of Planets and the Moon

Planet	Distance from Sun, 10^9 m	Calculated Temperature K	Actual Temperature K	ΔT
Venus	108	252	730	+478
Earth	150	255	288	+34
Earth's moon	150	270	274	+4
Mars	228	217	218	+1

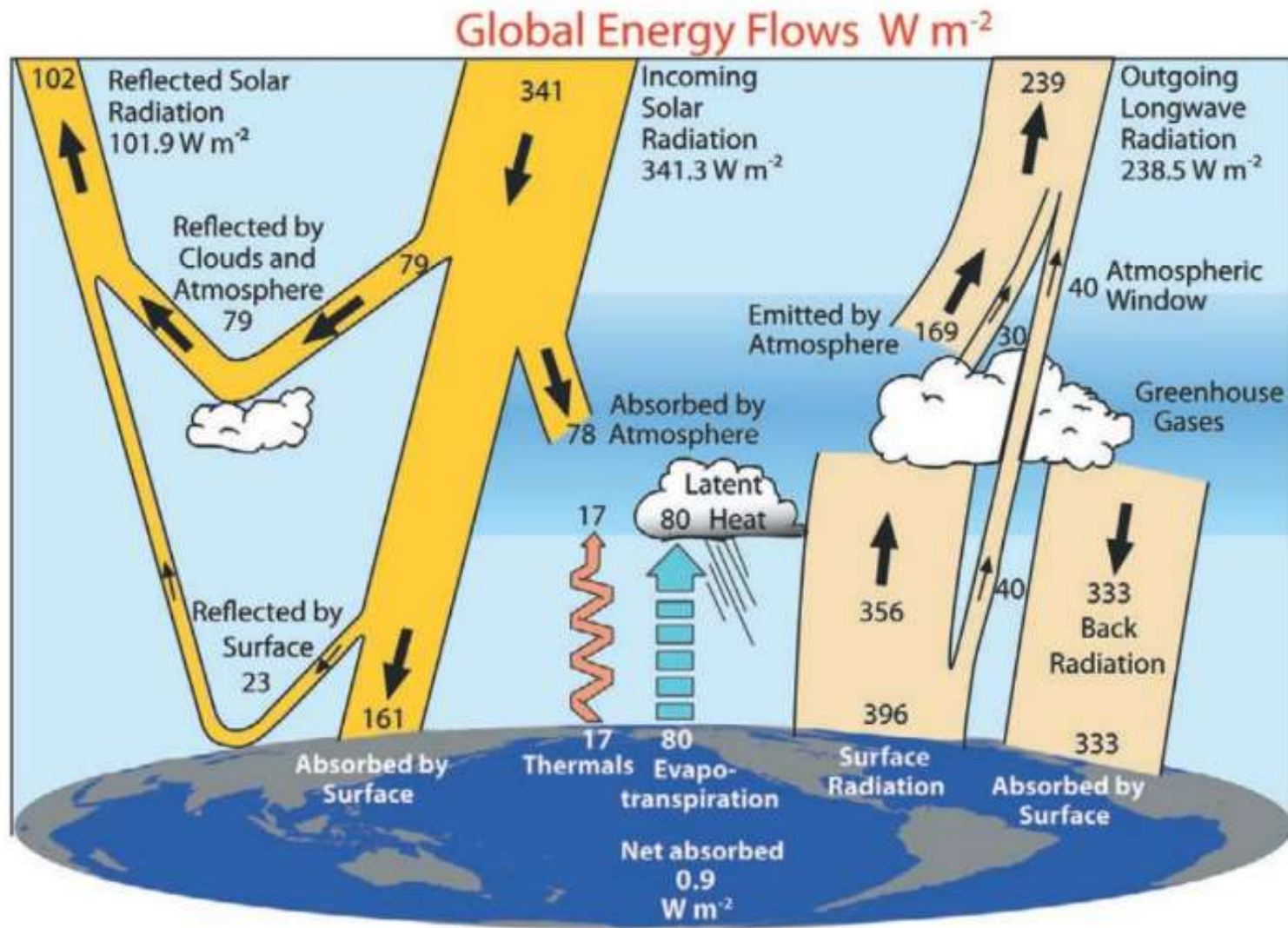
GREENHOUSE EFFECT: ABSORPTION OF TERRESTRIAL RADIATION BY THE ATMOSPHERE



Atmosphere is transparent to visible light

- **Major greenhouse gases: H_2O , CO_2 , CH_4 , O_3 , N_2O , CFCs, ... are gasses that absorb thermal infrared radiation**
- Not greenhouse gases: N_2 , O_2 , Ar, ...

WHY IS EARTH NOT 255 K COLD? BECAUSE OF THE NATURAL GREENHOUSE EFFECT!



WHAT IS A GREENHOUSE GAS?

A greenhouse gas (sometimes abbreviated GHG) is a gas that absorbs and emits radiant energy within the thermal infrared range. Greenhouse gases cause the greenhouse effect on planets.

The **primary greenhouse gas in Earth's atmosphere is water vapor (H_2O)**. The water vapour in the Earth's atmosphere is controlled by global sea surface temperature. Hence water vapour forcing is controlled by natural feedback cycles within in the earths climate system.

The most **important anthropogenic greenhouse gasses are carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and ozone (O_3)**. Without greenhouse gases, the average temperature of Earth's surface would be about $-18\text{ }^\circ\text{C}$ ($0\text{ }^\circ\text{F}$), rather than the present average of $15\text{ }^\circ\text{C}$ ($59\text{ }^\circ\text{F}$).

To act as climate changing greenhouse gas, a compound must absorb spectral lines that are not fully absorbed by water vapour already

RADIATIVE FORCING THE DEFINITION

Radiative forcing is the difference between insolation (sunlight) absorbed by the Earth and energy radiated back to space.

Numerically,

$$\Delta F = F_{\text{in}} - F_{\text{out}}$$

All changes to Earth's radiative equilibrium, that cause temperatures to rise or fall over decadal periods, are called climate forcings.

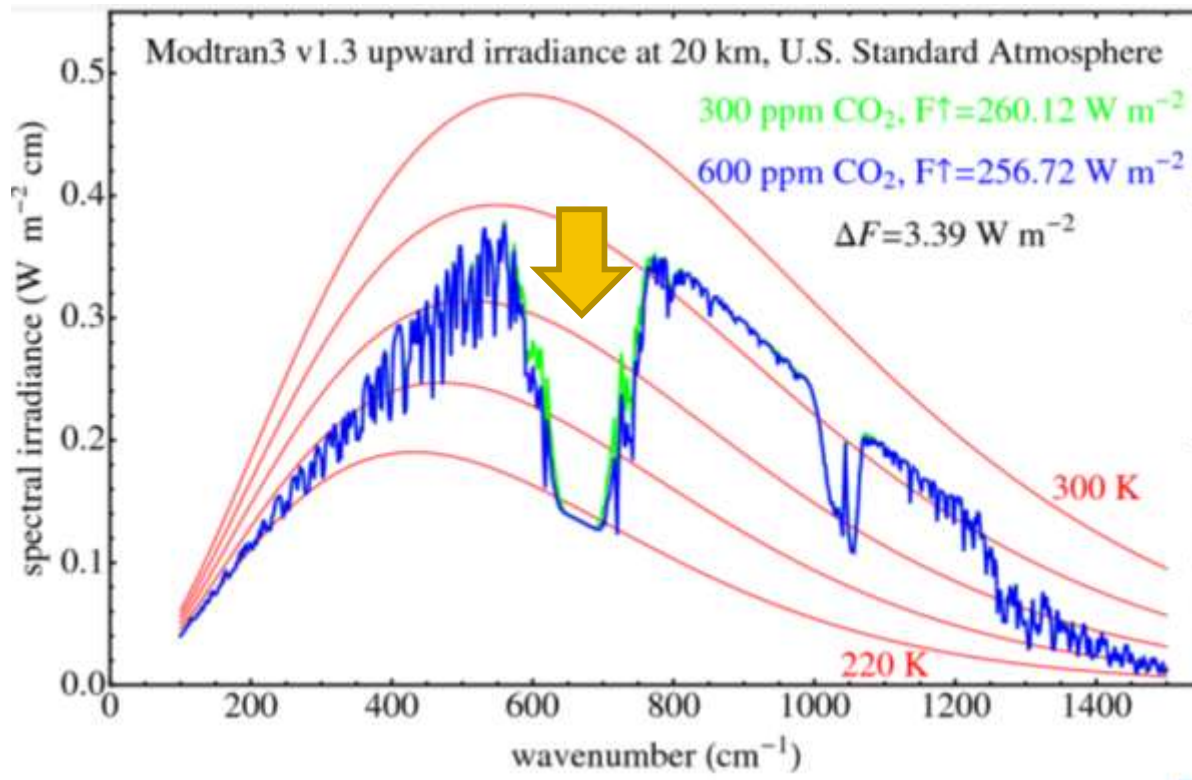
Positive radiative forcing means Earth receives more incoming energy from sunlight than it radiates to space. This net gain of energy will cause warming.

Conversely, negative radiative forcing means that Earth loses more energy to space than it receives from the sun, which produces cooling. A system in thermal equilibrium has zero radiative forcing.

•Radiative forcing due to a GHG X is defined as the flux imbalance in the radiative budget of the Earth system caused by adding a mass Δm of X to the atmospheric mass of the GHG X

ATTRIBUTION OF CHANGES IN FORCING TO CONCENTRATION CHANGES

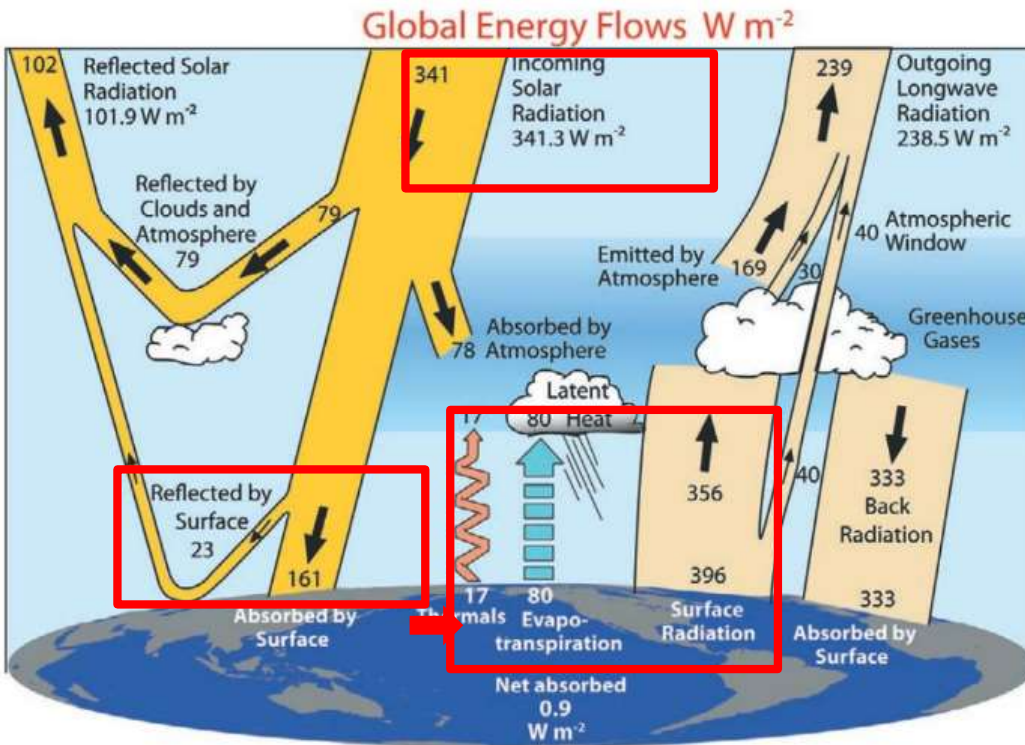
Atmospheric window



Radiative forcing for doubling CO₂, as calculated by radiative transfer code Modtran. Red lines are Planck curves.

When the Energy exiting the system changes because of concentration changes of greenhouse gasses then the only way to re-establish equilibrium is for the temperature of the black body called “Earth” to change its temperature to a different Plank curve.

PAST VARIATIONS IN RADIATIVE FORCING DEMONSTRATE THAT THE SYSTEM IS VERY SENSITIVE AND SMALL FORCING'S AMPLIFY



The incoming solar radiation of 341.3 W m^{-2} (annual average) varies only slightly as a consequence of variable solar activity and Earth's orbital parameters ($<1\%$)...

... but since the amount of radiation received during the summer months changes much more (by $\sim 8\%$) due to variations in the Earth's orbital parameters...

... and summer radiation determines where all snow will NOT melt, which changes the reflectance and absorption term small changes can be amplified!

Trenbeth et al. 2009, BAMS, DOI:10.1175/2008BAMS2634.1

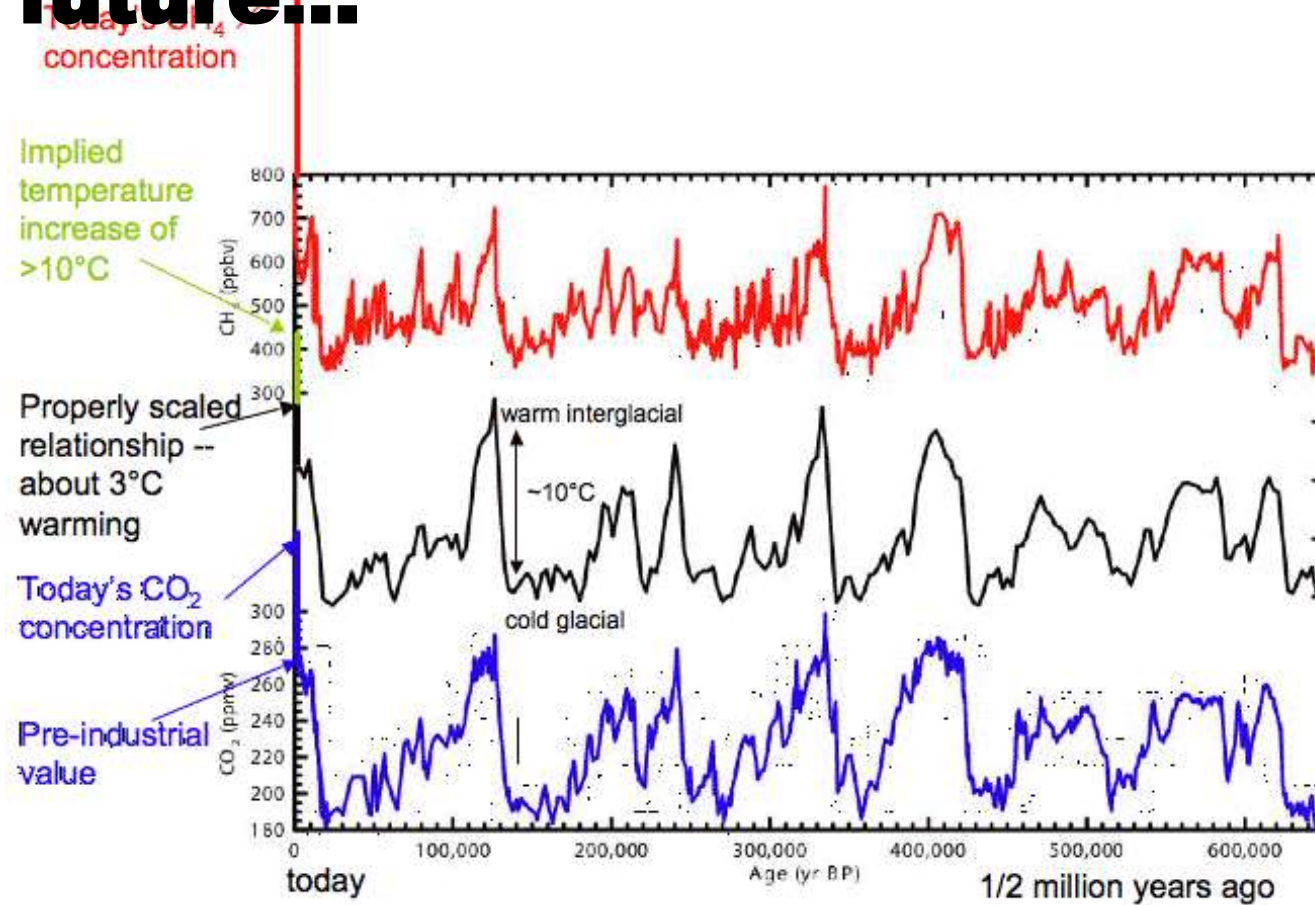
... also lower summer temperature means lower ocean temperature and less water vapour in the atmosphere.

WHAT ARE MILANKOVITCH (ORBITAL) CYCLES?

Variations in three types of Earth orbital movements affect how much solar radiation (known as insolation) reaches the top of Earth's atmosphere as well as where the insolation reaches. These cyclical orbital movements (eccentricity, axis tilt and precession), which became known as the Milankovitch cycles, cause variations of up to 25 percent in the amount of incoming summer insolation at Earth's mid-latitudes (the areas of our planet located between about 30 and 60 degrees north and south of the equator).

The response of the earth system to these perturbations can teach us a lot about the climate sensitivity of the Earth system

The past is used to be the key to the future...



Methane, **temperature** (from hydrogen isotope ratios ("δD") and **carbon dioxide** from the Dome C Ice core. (EPICA Project members, 2006).

Whenever there is more summer time radiation ice melts, resulting in more absorption of solar radiation and warming of the oceans, which release CO₂ into the atmosphere.

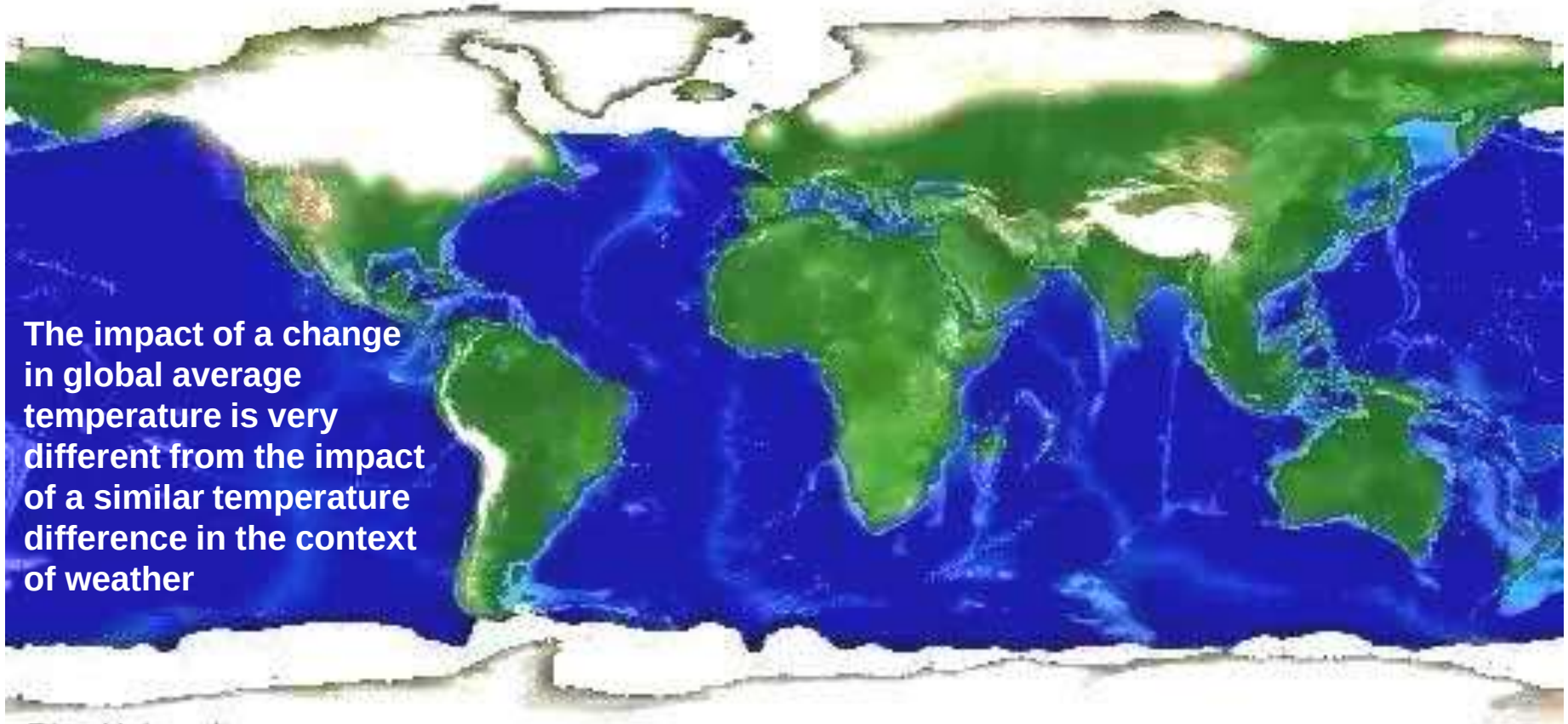
Methane production of terrestrial ecosystems increases with temperature, and melting permafrost releases additional methane

Steig, Univ. Washington.

Source: <http://www.ess.washington.edu/~steig/images/epicagore.gif>

Forcing of -12 W/m^2 during summer and -8 to -8.5 W/m^2 annual average (=annual average temperatures 4.3 to 9.8°C below the Holocene average) @ $18,000 \text{ BP}$ made the planet look like this:

What did it take for all this to start melting? $+32 \text{ W/m}^2$ summer, -0.1 W/m^2 annual average @ $10,000 \text{ BP}$ triggered the melting
2019 forcing: $+ 2.72 \text{ W/m}^2$



The impact of a change in global average temperature is very different from the impact of a similar temperature difference in the context of weather

Rice University

<http://water.usgs.gov/edu/watercycleice.html>

GEOPHYSICAL RESEARCH LETTERS, VOL. 33, L14709,

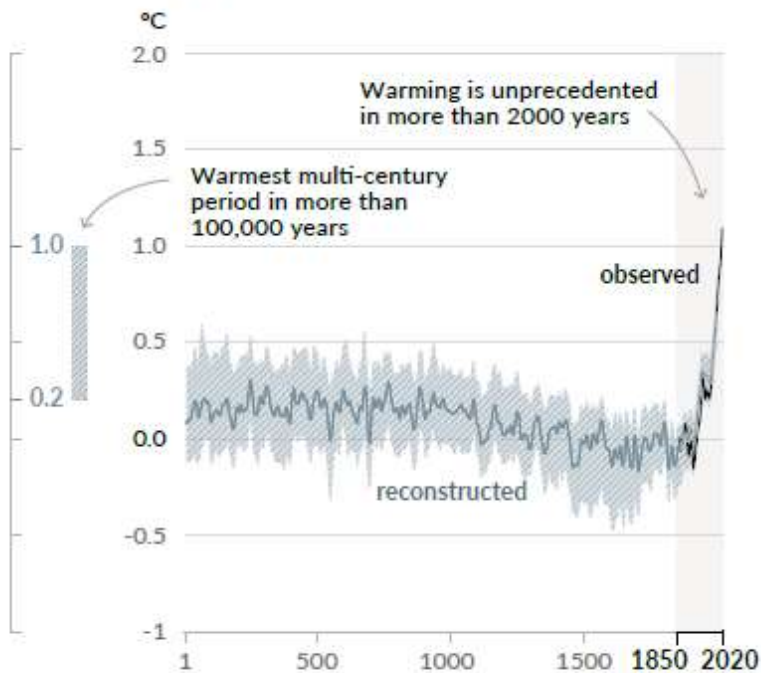
doi:10.1029/2006GL026484, 2006

Otto-Bliesner et al. 2005 Journal of climate, IPCC 2014

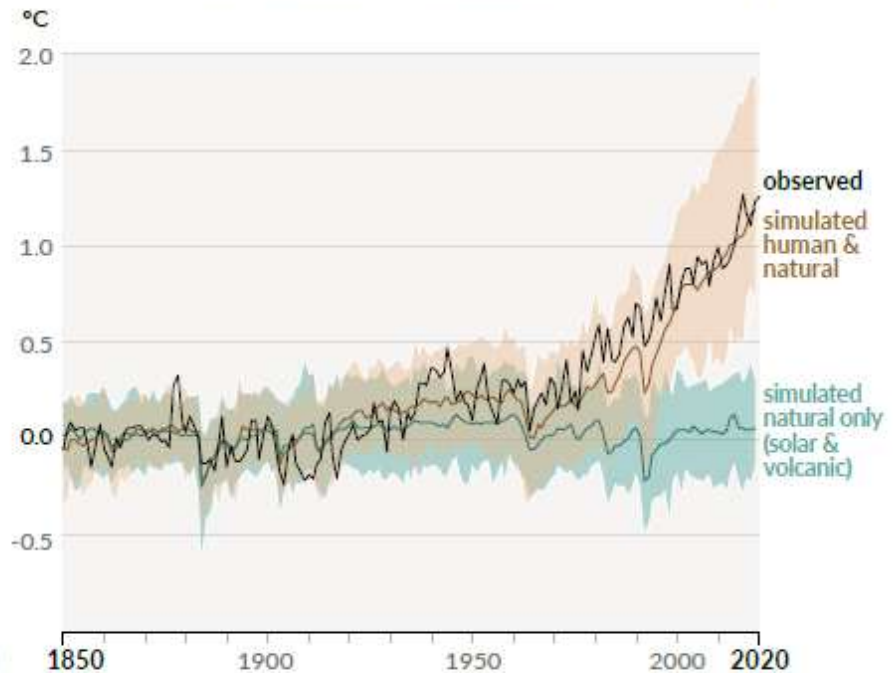
ARE MILANKOVITCH CYCLES RESPONSIBLE FOR THE CURRENT WARMING?

Changes in global surface temperature relative to 1850-1900

a) Change in global surface temperature (decadal average) as reconstructed (1-2000) and observed (1850-2020)



b) Change in global surface temperature (annual average) as observed and simulated using human & natural and only natural factors (both 1850-2020)

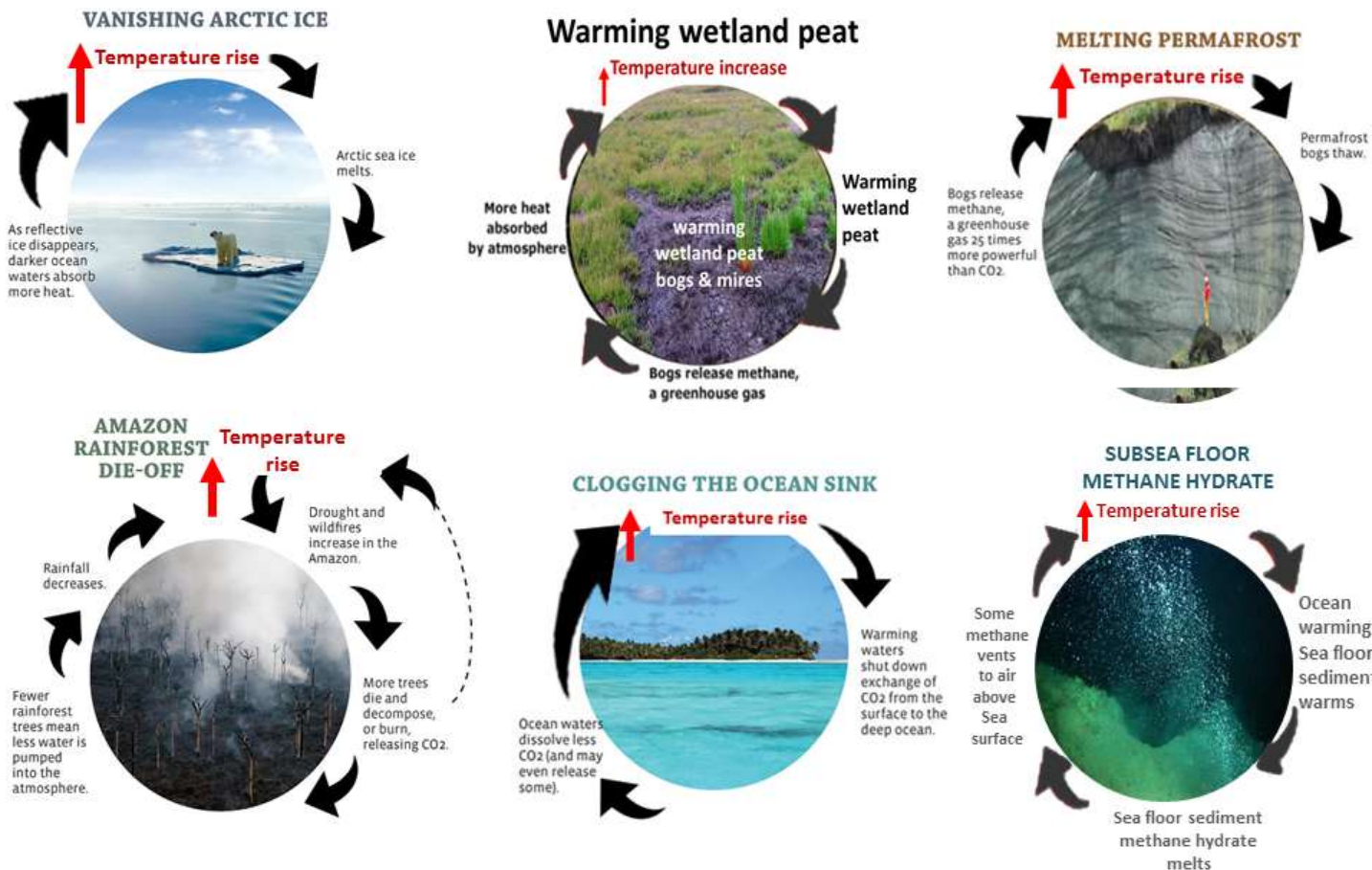


Based on natural forcing's the next thing on our geological calendar was the next ice age. Only human emissions explain the recent warming

What makes the system so unstable?

Some large +ve amplifying feedbacks

**POSITIVE
FEEDBACK
CYCLES ARE
CYCLES WHERE A
SMALL CHANGE IN
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TRIGGERS A
SYSTEM
RESPONSE THAT
FORCES THE
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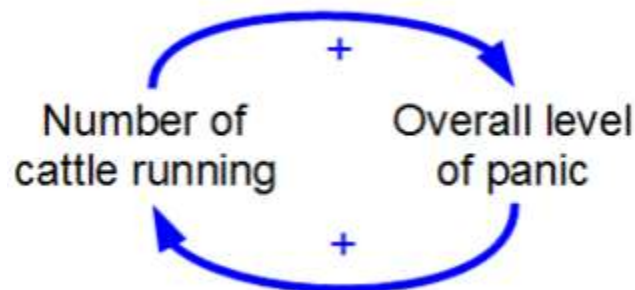
In the context of climate change positive feedback cycles are neither good nor helpful. Due to them the climate system is fragile and easily disturbed by small changes.

http://www.climateemergencyinstitute.com/uploads/six_large_feedbacks.png

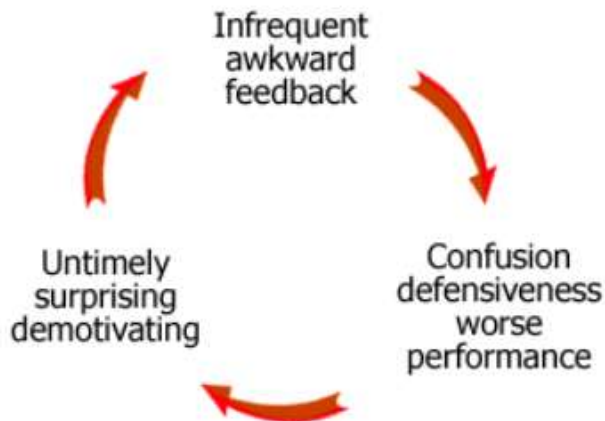
Positive feedback cycles make the system unstable

POSITIVE FEEDBACK CYCLES ARE CYCLES WHERE A SMALL CHANGE IN ONE DIRECTION, TRIGGERS A SYSTEM RESPONSE THAT FORCES THE SYSTEM FURTHER INTO THE SAME DIRECTION

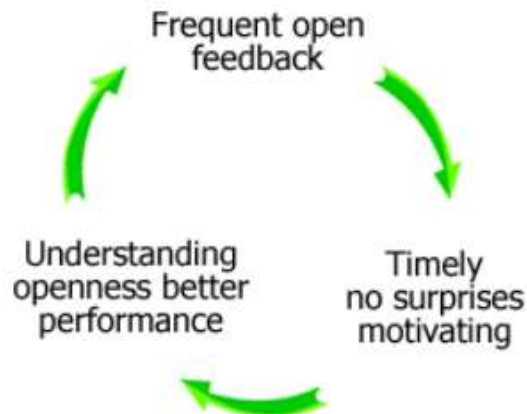
Stampede feedback loop



Vicious Circle



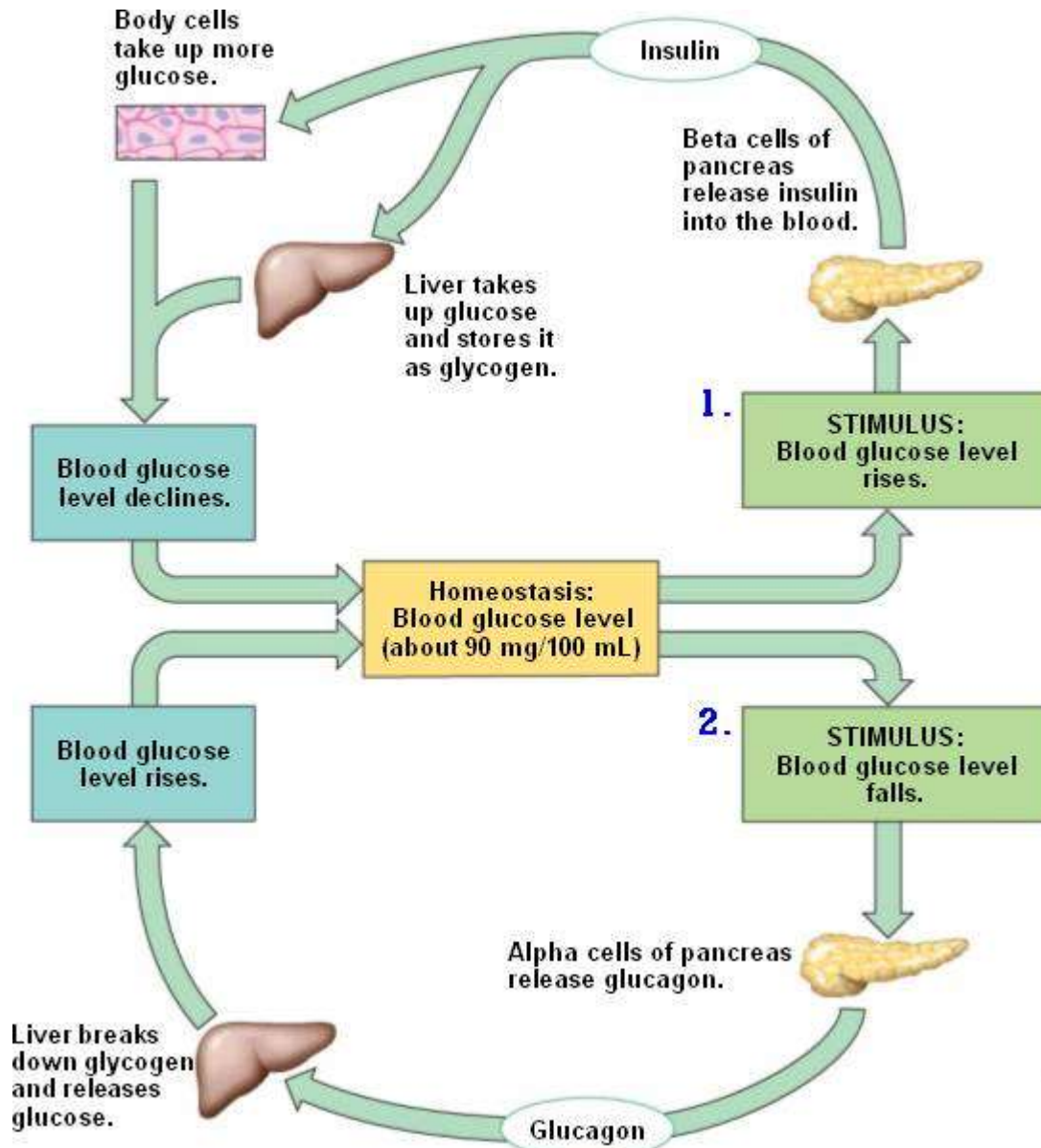
Virtuous Circle



In the context of climate change positive feedback cycles are neither good nor helpful. Due to them the climate system is fragile and easily disturbed by small changes.

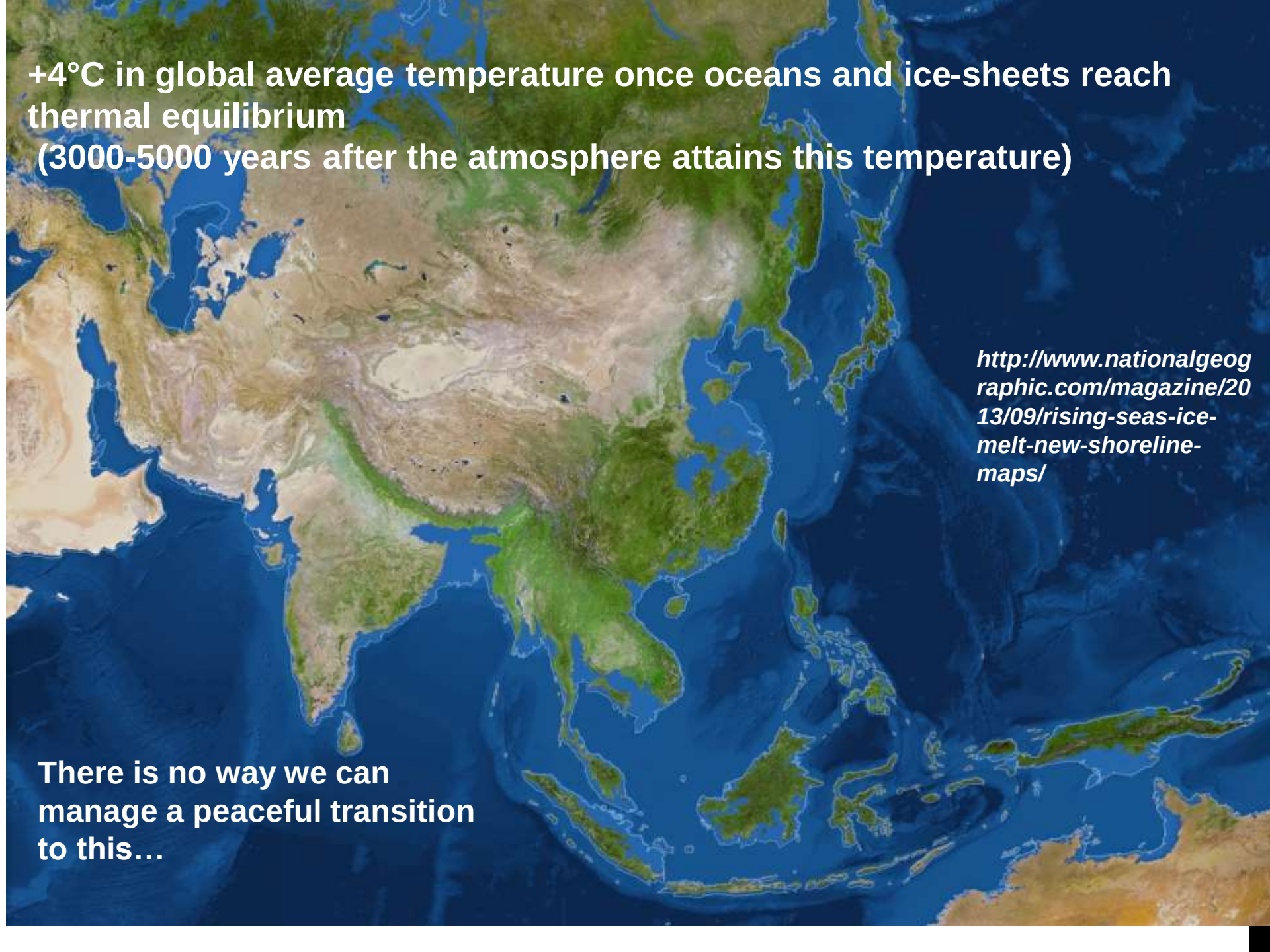
Negative feedback cycles stabilize the system

Blood sugar regulation in the body



NEGATIVE FEEDBACK CYCLES ARE CYCLES WHERE A SMALL CHANGE IN ONE DIRECTION TRIGGERS A SYSTEM RESPONSE THAT REIGNS IN THE STIMULUS AND RESTORS THE ORIGINAL STATE OF THE SYSTEM ...

In the context of climate change negative feedback cycles act very slowly and take millennia during which the planet is in a hot and difficult to survive state to restore the original system state

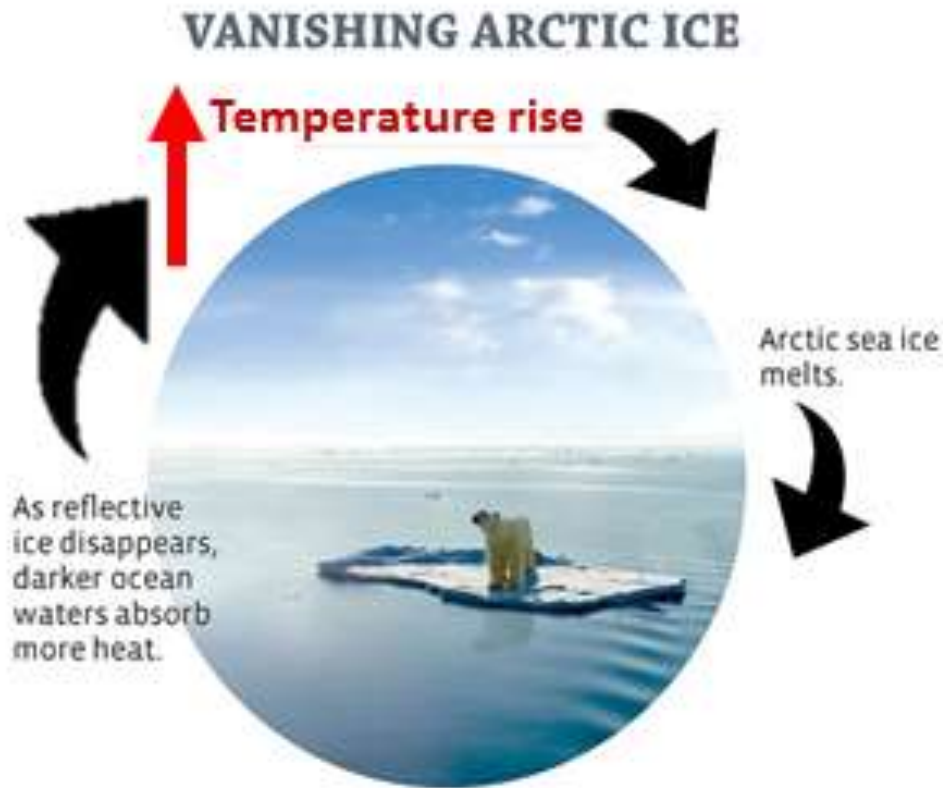


**+4°C in global average temperature once oceans and ice-sheets reach thermal equilibrium
(3000-5000 years after the atmosphere attains this temperature)**

<http://www.nationalgeographic.com/magazine/2013/09/rising-seas-ice-melt-new-shoreline-maps/>

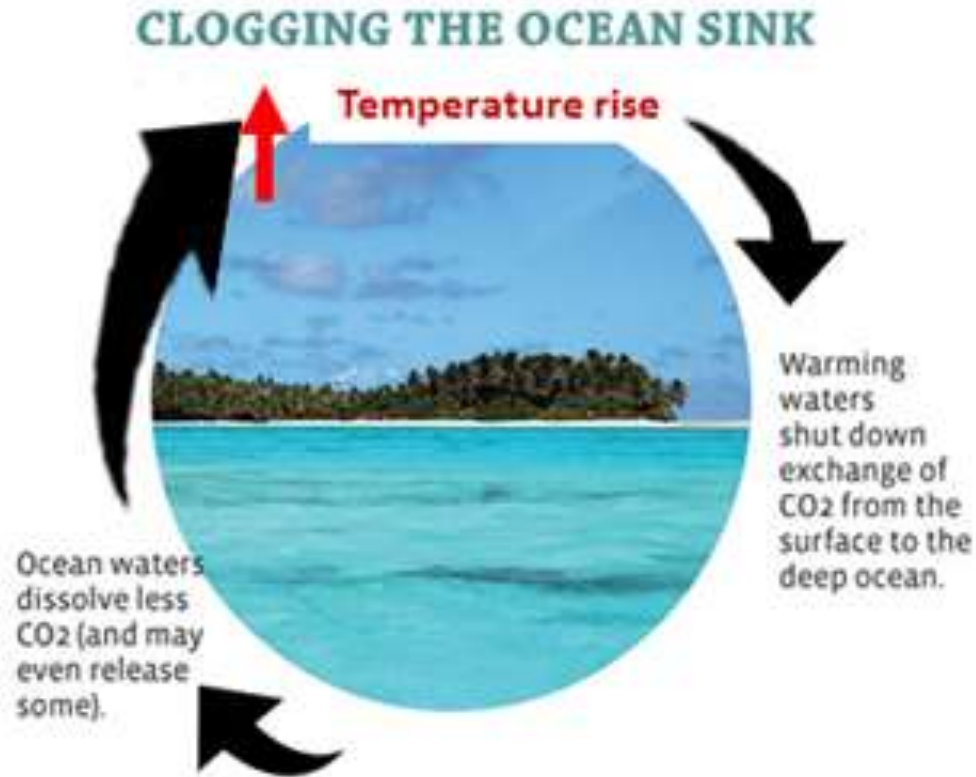
**There is no way we can
manage a peaceful transition
to this...**

Climate feedback cycles



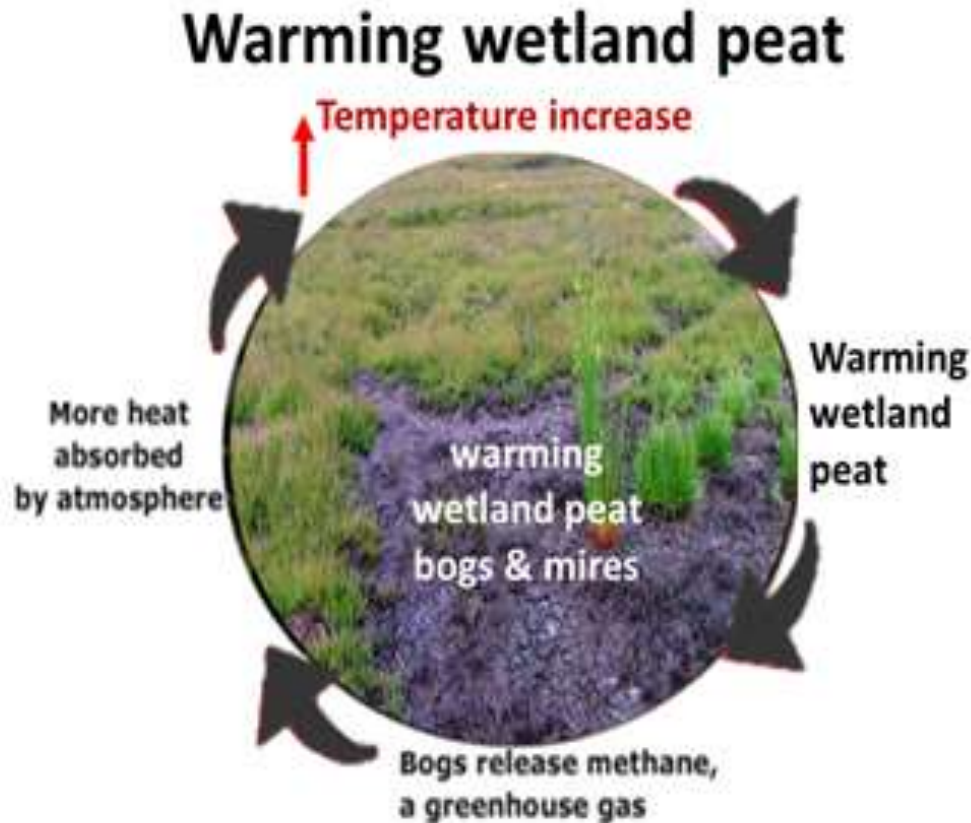
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Climate feedback cycles

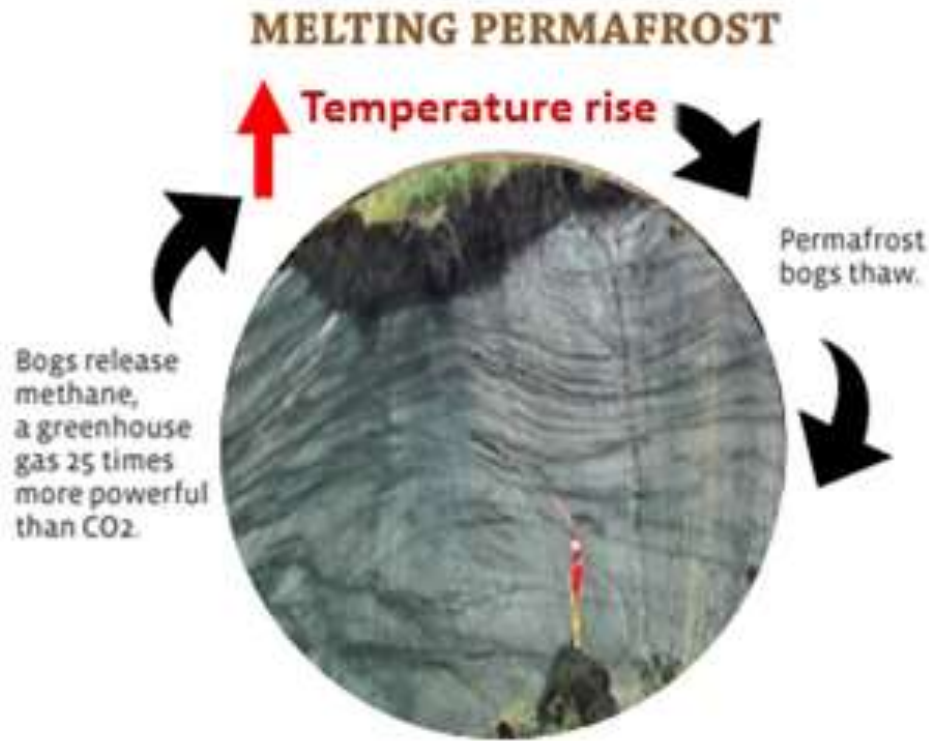


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Climate feedback cycles



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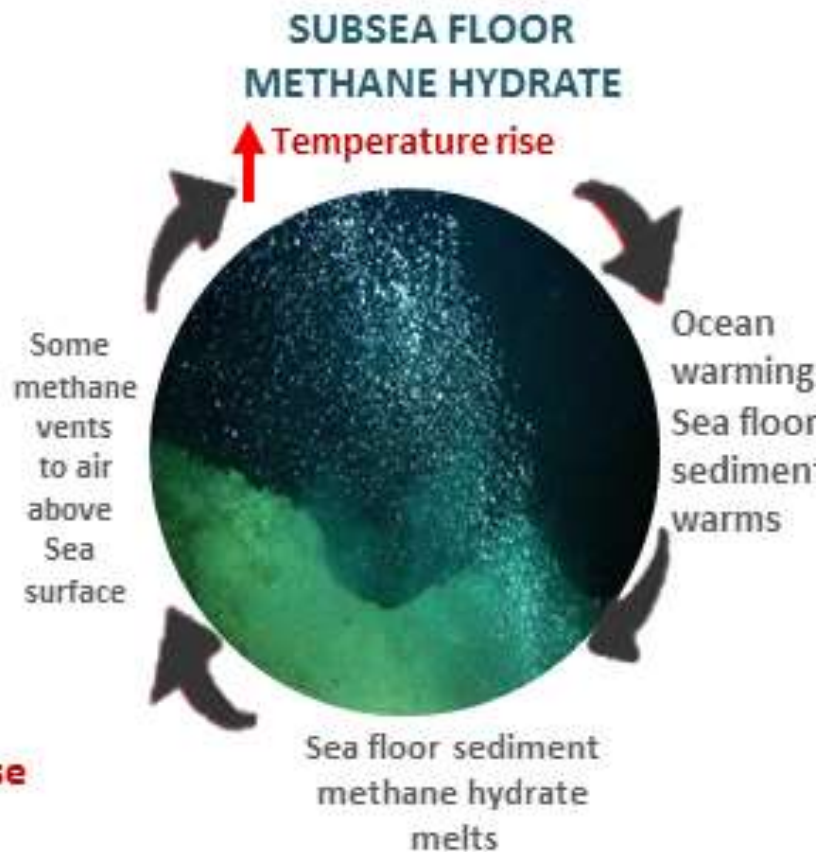
Climate feedback cycles

The climate wildcard – extreme runaway greenhouse effect

Last time the methane hydrates melted at a large scale, climate warmed by $>12^{\circ}\text{C}$ in a very short time and it took more than 200,000 years for methane levels to come back too normal

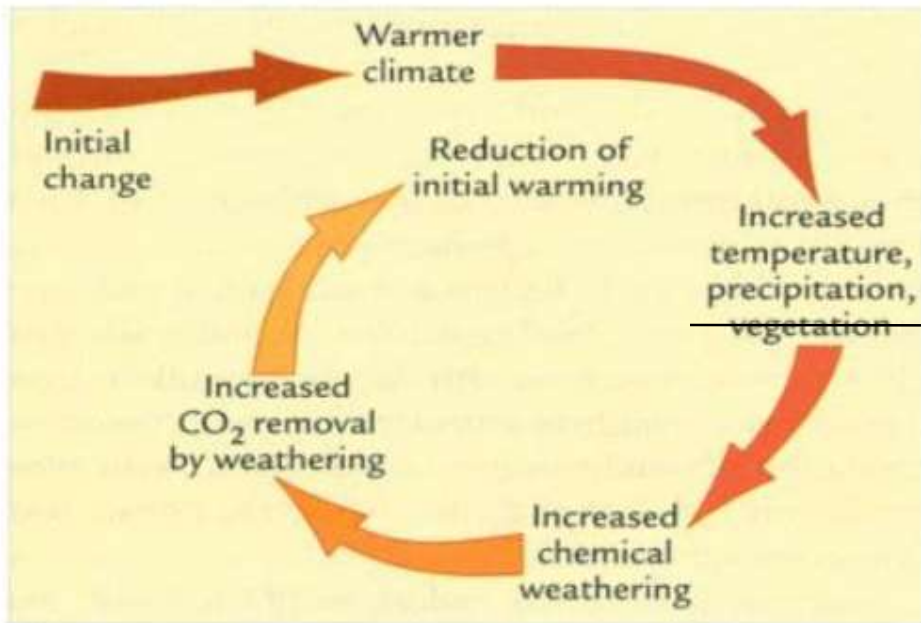
This methane peaks likely happened in the Paleocene Eocene thermal maximum PETM & in the Cretaceous when there was rapid sea floor spreading and the formation of the Deccan traps occurred and the position of continents was different. Consequently global ocean currents worked differently.

We don't understand how likely this is to happen and at what ocean temperature



http://www.climateemergencyinstitute.com/uploads/six_large_feedbacks.png

Climate feedback cycles: negative



This transfers CO₂ into carbonate rocks
 The accelerated hydrological cycle and the accompanying erosion and sedimentation also buries organic carbon

